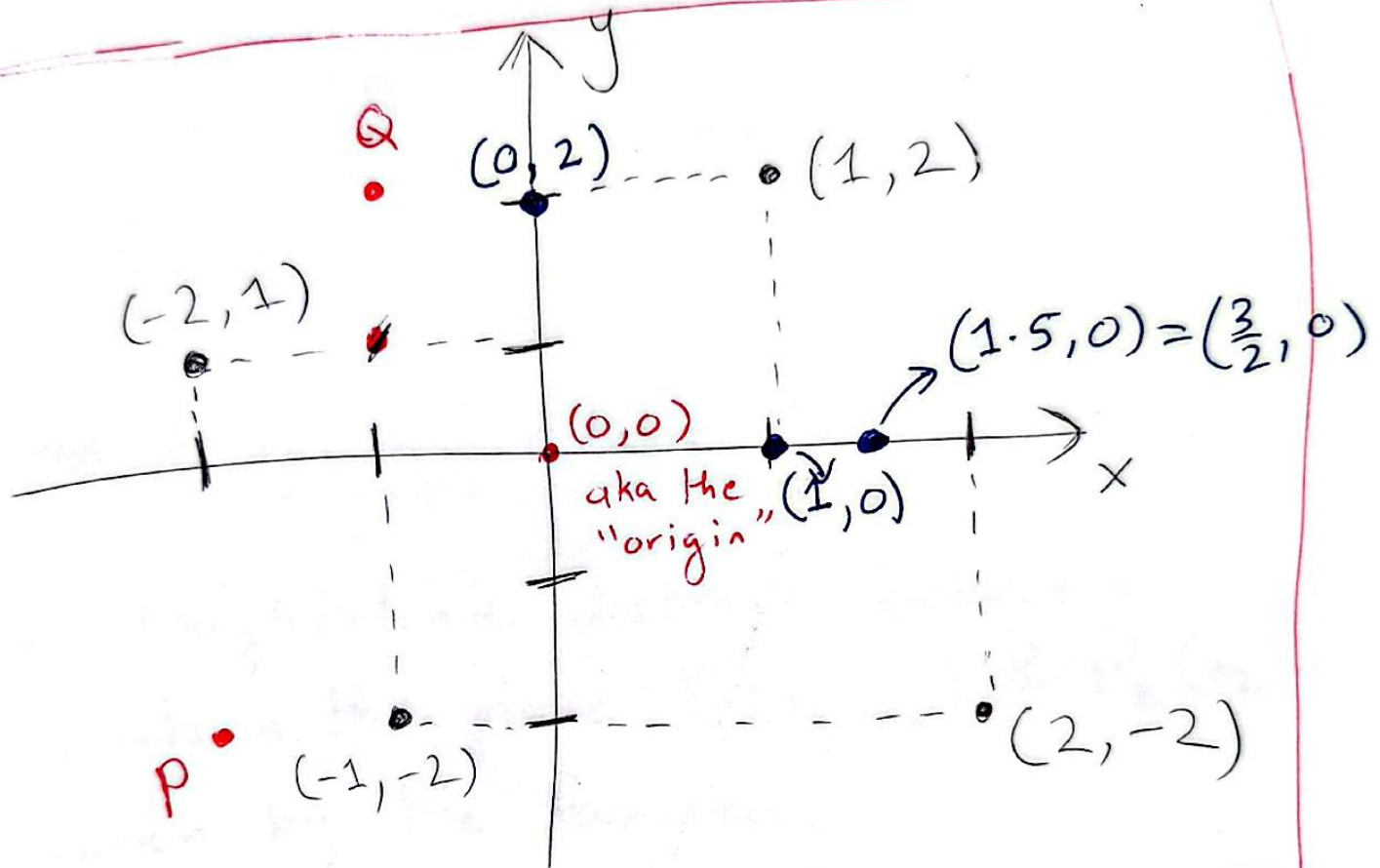


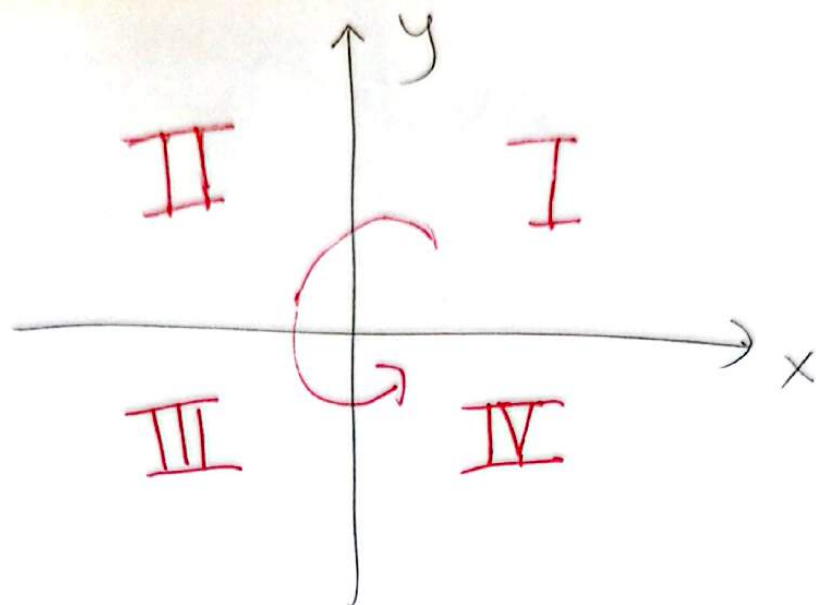
Rectangular Coordinate Systems



Q. What are the blue points?

$(0,2)$, $(1,0)$, $(1.5,0)$

Q. What does the ~~stat~~ expression $P(-2,-2)$ mean? $Q(-1,2)$?

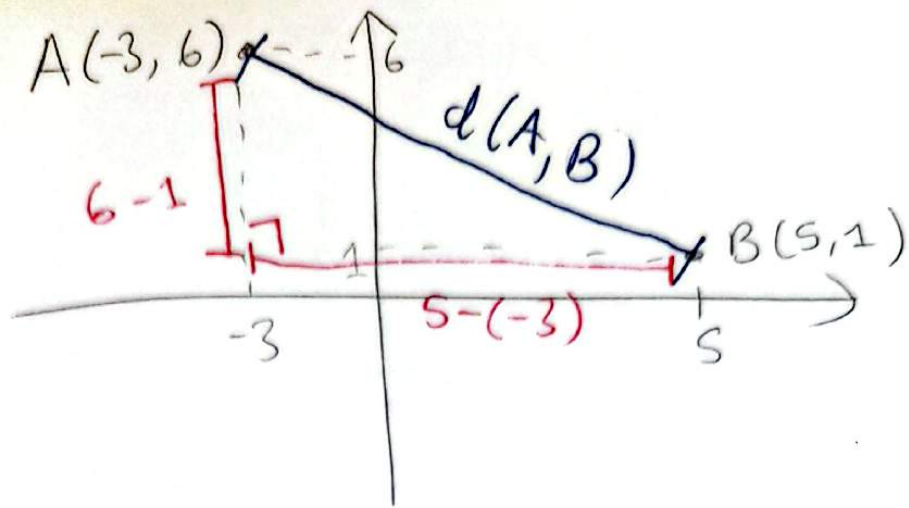


Distance in xy-plane:

The straight-line distance between two points in the plane $P_1(x_1, y_1)$ & $P_2(x_2, y_2)$ is given by the formula:

$$\begin{aligned} d(P_1, P_2) &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \end{aligned}$$

(why does this formula work?)



Find $d(A, B)$

$$d(A, B) = \sqrt{(5 - (-3))^2 + (1 - 6)^2}$$

$$= \sqrt{8^2 + (-5)^2}$$

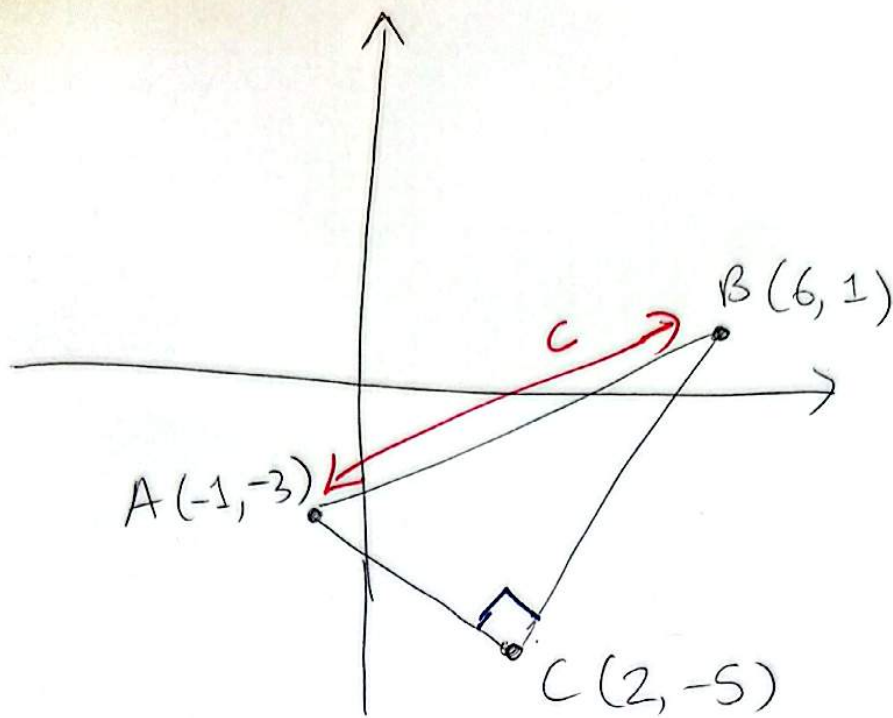
$$= \sqrt{64 + 25}$$

$$= \sqrt{89}$$

$$\sqrt{((-3) - 5)^2 + (6 - 1)^2}$$

$$= \sqrt{(-8)^2 + 5^2}$$

$$= \sqrt{89}$$



Show that this is a right-angled triangle.

LOGIC: Find lengths, plug into Pythagorean formula, and check

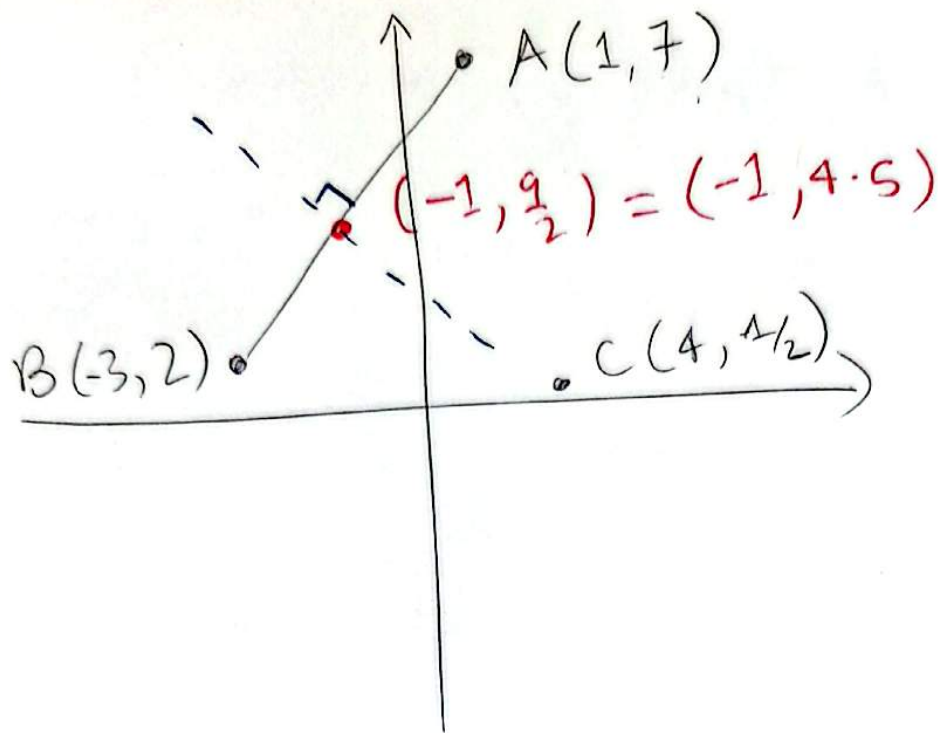
$$d(A, B) = \sqrt{(6 - (-1))^2 + (1 - (-3))^2} = \sqrt{65} \rightarrow \text{hypotenuse} = c$$

$$d(A, C) = \sqrt{(2 - (-1))^2 + (-5 - (-3))^2} = \sqrt{13} \rightarrow a$$

$$d(B, C) = \sqrt{(6 - 2)^2 + (1 - (-5))^2} = \sqrt{52} \rightarrow b$$

$$(\sqrt{13})^2 + (\sqrt{52})^2 = (\sqrt{65})^2$$

$$13 + 52 = 65 \quad \checkmark$$



What's the midpoint of A and B?

MP of $P_1(x_1, y_1)$ and

$$P_2(x_2, y_2) = MP\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$\begin{aligned} \text{MP of A \& B} &= \left(\frac{1 + (-3)}{2}, \frac{7 + 2}{2}\right) \\ &= \left(-1, \frac{9}{2}\right) \end{aligned}$$

How can we verify that this is a true midpoint?

Let's at least check it is "equidistant" to A and B.

$$A(1, 7), B(-3, 2), MP(-1, \frac{9}{2})$$

$$\begin{aligned} d(A, MP) &= \sqrt{(1 - (-1))^2 + (7 - \frac{9}{2})^2} \\ &= \sqrt{4 + \frac{25}{4}} = \frac{\sqrt{41}}{2} = \sqrt{\frac{41}{4}} \end{aligned}$$

$$\begin{aligned} d(B, MP) &= \sqrt{(-1 - (-3))^2 + (\frac{9}{2} - 2)^2} \\ &= \sqrt{4 + \frac{25}{4}} = \frac{\sqrt{41}}{2} = \sqrt{\frac{41}{4}} \end{aligned}$$

Check that $C(4, \frac{1}{2})$ is on the perpendicular bisector of $A(1, 7)$ and $B(-3, 2)$.

$$d(A, C) \stackrel{?}{=} d(B, C)$$

$$d(A, C) = \sqrt{(4-1)^2 + (7-\frac{1}{2})^2}$$
$$= \sqrt{9 + \frac{169}{4}}$$

$$\frac{1}{2} \cdot \sqrt{205}$$

$$= \sqrt{\frac{36}{4} + \frac{169}{4}} = \left(\frac{\sqrt{205}}{2} \right) = \sqrt{\frac{205}{4}}$$

$$d(B, C) = \sqrt{(4-(-3))^2 + (2-\frac{1}{2})^2}$$

$$= \sqrt{7^2 + (\frac{3}{2})^2}$$

$$= \sqrt{49 + \frac{9}{4}}$$

$$= \sqrt{\frac{196+9}{4}} = \sqrt{\frac{205}{4}} = \left(\frac{\sqrt{205}}{2} \right)$$

can we find a general equation in x and y for the points on the perp. bisector of $A(1, 7)$ & $B(-3, 2)$?

Let's say $P(x, y)$ is on the perp. bisector

$$\iff d(A, P) = d(B, P)$$

$$\iff \sqrt{(x-1)^2 + (y-7)^2} = \sqrt{(x-(-3))^2 + (y-2)^2}$$

↓ squaring both sides

$$\iff \underbrace{(x-1)^2}_{\downarrow} + \underbrace{(y-7)^2}_{\downarrow} = \underbrace{(x+3)^2}_{\downarrow} + \underbrace{(y-2)^2}_{\downarrow}$$
$$\iff \cancel{x^2} - 2x + 1 + \cancel{y^2} - 14y + 49 = \cancel{x^2} + 6x + 9 + \cancel{y^2} - 4y + 4$$

$$\iff -2x - 6x - 14y + 4y + 1 + 49 - 9 - 4 = 0$$

$$\iff -8x - 10y + 37 = 0$$

$$\boxed{-8x - 10y = -37} \text{ STANDARD FORM}$$

$$-8x - 10y = \cancel{37} - 37$$

$$\Rightarrow -10y = -37 + 8x$$

$$\Rightarrow y = -\frac{8}{10}x + \frac{37}{10}$$

$$\boxed{y = -\frac{4}{5}x + \frac{37}{10}} \rightarrow \text{slope-intercept form}$$

$$ax + by = c$$

$$y = mx + c$$